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by Y A

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Sunlike full-spectrum quantum dots (QDs) based on I–III–VI group semiconductors are promising candidates for next-generation, cadmium-free optoelectronic applications. They are perfect for warm-white and sunlike light-emitting devices because of their innately broad emission, configurable bandgap, low toxicity, and defect-tolerant electrical structure. Nevertheless, it is still difficult to achieve high photoluminescence quantum yield (PLQY), spectrum homogeneity, and device stability at the same time. This study merges first-principles computational design, optimal synthetic methods, and practical QLED fabrication to establish structure–property–device correlations and provide high-performance full-spectrum QLEDs employing I–III–VI QDs. We intend to cover the size and defect engineering, underlying mechanics of emission, and optimization of the newly designed device (s) for high-efficiency, steady emission.

The main objective of the research is to look for alternate energy solutions by designing and synthesizing the materials with optoelectronic properties. I-III-VI QDs belong to the family of quantum dots that comprise elements from Groups I, III, and VI of the periodic table. It implies that heavy toxic elements like cadmium are not present in them. They are favored because they are environmentally friendly QD technology. The versatility of these QDs is explained by their broad tunability for a wide range of emission spanning from the visible to the near infrared region. I-III-VI QDs may find application in full-spectrum light sources such as QLEDs, allowing them to be used in displays and lighting solutions. These QDs can be fabricated into multilayer device architecture to enable the realization of their practical applications.

This project has its significant impact in many instances:

1) The project is expected to contribute in global efforts for sustainable energy materials.

2) Cd- and Pb-free quantum dots keeping clean and green energy in mind.

3) The project will be started with the intention of lab to industry pathway. Device fabrication will help in testing the industry-level performance.

The research addresses various challenges such as:

1) Getting rid of toxic cadmium and lead based quantum dots.

2) Computational design and simulations before exhaustive laboratory exercises.

3) Material properties to be reproduced in real-world device.

The selection of chemicals and solvents suitable for the synthesis of I–III–VI quantum dots is the first step in the entire experimental approach for this process. Device fabrication and evaluation of QLED devices composed of these materials come next. The main group I

precursor is copper (I) iodide or copper (I) acetate; group III metals can be obtained by supplementing indium acetate with gallium acetylacetonate. A chalcogen source can be provided by dissolving sulfur powder in oleylamine or selenium in trioctylphosphine. The whole synthesis method is carried out in a colloidal environment with oleylamine, oleic acid, and 1-octadecene serving as solvents, coordinating ligands, and stabilizing agents at the same time. The overall approach combines computational modeling, device fabrication, and quantum dot synthesis. To enable ongoing optimization of quantum dots and their corresponding electroluminescent devices, each step is connected to the others.

2. I–III–VI Quantum Dot Synthesis

The modified hot-injection colloidal method of synthesis, which provides excellent control over nucleation and growth dynamics, is used for obtaining full-spectrum I-III-VI quantum dots similar to sunlight. The first step includes the dissolution of the metal precursors in the three-neck flask containing oleylamine and octadecene. In order to eliminate any residual moisture and dissolved oxygen to provide a clean reaction system, this mixture is subjected to the vacuum heating up to 120 °C. In addition, another solution of chalcogen precursor is prepared. For the synthesis of sulfur-based quantum dots, elemental sulfur is dissolved in oleylamine at the temperature ranging from 80 to 100 °C until a transparent solution appears. In selenium-containing compositions, a reactive TOP-Se complex is formed from selenium powder dissolved in trioctylphosphine. After the preparation of the precursors, the metal precursor solution is gradually heated up to the reaction temperature, which ranges from 180 to 220 °C, under nitrogen atmosphere. At this point, the heated metal precursor solution is rapidly added to the chalcogen precursor. This leads to the controlled process of crystallization after the rapid nucleation. The emission wavelength of the quantum dots may be varied throughout the whole visible spectrum depending on the growth time and reaction temperature. Short reaction times give smaller particles with greenish emission, while long growth times provide larger nanocrystals with orange, yellow, or red emission. Heating mantle is taken off, and the temperature flask is cooled in the water bath in order to stop the reaction process.

3. Ligand Modification and Post-Synthesis Surface Engineering

Following purification, the quantum dots undergo ligand engineering to increase their optoelectronic efficiency and suitability for QLED device designs. When incorporated into electroluminescent devices, the long-chain ligands used in synthesis restrict charge injection and transfer despite their high colloidal stability. Therefore, shorter-chain ligands, such as octylamine or thiolbased ligands, are introduced into the quantum-dot dispersion using a ligand exchange technique. These shorter ligands replace the main long-chain molecules, resulting in films with balanced charge transfer and increased conductivity. To

increase photoluminescence quantum yield and durability, especially for broad-spectrum white emission, a ZnS shell can be created around the quantum dot core. Zinc and sulfur precursors are injected dropwise into the core nanocrystals after they have been gradually heated to a moderate temperature.

4. Optical, Structural, and Electronic Characterization

A complete characterization approach is used to determine the optical, structural, and electrical characteristics of the synthesized quantum dots. Band gaps and size-dependent optical transitions are measured using UV-visible absorption spectroscopy. Fullspectrum emission is confirmed and emission profiles are provided by photoluminescence spectroscopy. Radiative efficiency is estimated by photoluminescence quantum yield evaluations using an integrating sphere. To investigate exciton dynamics and distinguish between radiative and non-radiative mechanisms, time-resolved photoluminescence experiments are employed. Colorimetric analysis, which includes correlated color temperature (CCT), color rendering index (CRI), and CIE coordinates, is used to evaluate the regularity and quality of full-spectrum emission.

5. Theoretical analysis and computational modeling

The process includes computer modeling in addition to experimental work. Band structures, state density, and the impact of composition ratio changes on electronic transitions are estimated using Density Functional Theory (DFT) simulations. Depending on the computational resources available, software applications like Quantum ESPRESSO, VASP, and Gaussian are used to run simulations. Additionally, different surface ligands' reactions with quantum-dot surfaces are studied using ligand-binding calculations for energy and molecular dynamics simulations. This aids in identifying the ligand complexes that will result in higher stability and charge transfer. Optical simulations using time-dependent DFT (TD-DFT) can assist in predicting the absorption and emission properties of core and core-shell structures. Faster optimization and less trial-and-error testing are achieved by using computational insights to support experimental modifications of precursor quantities, shell thickness, and ligand selection.

6. Manufacturing QLED Devices

The first stage in creating quantum-dot light-emitting diodes is the creation of glass substrates coated with ITO. Before being treated with UV-ozone, these substrates are ultrasonically rinsed with detergent, deionized water, acetone, and isopropanol to increase surface wettability. A thin layer of PEDOT:PSS is spin-coated onto the substrate as the hole-injection layer, and it is then thermally baked to remove moisture. Next, PVK or Poly-TPD spin-coating is used to deposit the hole-transport layer. The emissive layer is produced by

spin-coating a dispersion of artificial quantum dots in toluene. This emissive layer may consist of a highly optimized combination of blue-, green-, and red-emitting I-III-VI quantum dots or a single type of broadband quantum dot that emits like the sun, depending on the technique.

The current project will be beneficial with multidimensional outcomes. Below is the summary of our expected results which signifies the immediacy of the project:

1) Energy Harvesting: In the ever-increasing population era, the scientific community is after sustainable energy solutions. I-III-VI quantum dots with sunlike full spectrum emission are potentially a versatile option for exploring their applications in energy storage devices such as batteries, solar panels, and other semiconducting devices.

2) Clean and green QD Technology: Following the 7th sustainable development goal of the UN, the current project will be a valuable contribution to synthesizing cadmium- and lead-free I-III-VI QDs.

3) Device Fabrication: The newly synthesized QDs will be tested for their performance by employing them in QLED devices.

4) Controlled and Efficient Synthesis of QDs: The synthesis of I-III-VI quantum dots and their tailoring for desired results will be powered with computational support. Consequently, a controlled synthesis is expected as the optical and thermoelectric properties will be computationally optimized before starting the wet-lab procedures.

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